



IETF-126 Hackathon

IPWASN Side Meeting



IPv6 Wireless Access in Satellite Networks (IPWASN) Project

July 22, 2026, Vienna

Champion: Jaehoon (Paul) Jeong

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IETF-126 IPv6 Wireless Access in Satellite Networks (IPWASN)

Champion: Jaehoon Paul Jeong (SKKU)



IPv6 Wireless Access in Satellite Networks (IPWASN) Project

Professors:

- Jaehoon Paul Jeong (SKKU)
- Takuya Yoshihiro (Wakayama Univ.)
- Tae (Tom) Oh (RIT)
- Yiwen (Chris) Shen (Ajou Univ.)
- Xiaohong (Dawn) Yu (Ajou Univ.)

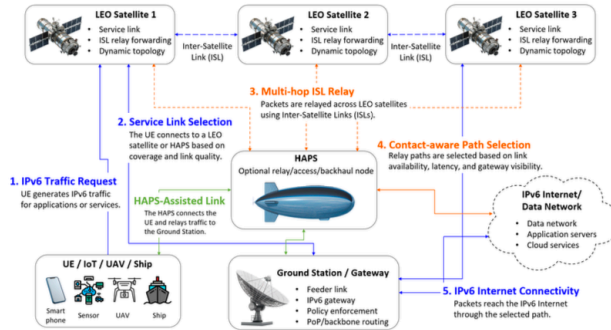
Researchers:

- Jung-Soo Park (ETRI)
- Byoungman Robert An (KETI)
- Bien Aime Mugabarigira (SKKU)

Students:

- Jimin Lee (SKKU)
- Eunjin Hwang (SKKU)
- Xudong Wang (SKKU)
- Shota Okazaki (Wakayama Univ.)

IPWASN Reference Scenario



Main Goals

- To combine IPWASN with prediction-based forwarding concepts from data forwarding schemes (e.g., TPD and TSF).
- To predict forwarding opportunities using satellite ephemeris, HAPS trajectory, gateway visibility, and link availability.
- To evaluate delivery delay, delivery ratio, handover disruption, and path stability through simulation.
- To analyze SRv6/SR-MPLS support for prediction-assisted traffic engineering and fast reroute.
- To design QoS-aware path selection for hybrid NTN paths.
- To prepare prototype support for validating IPWASN requirements and future protocol extensions.

IPWASN Developing Environment

- Simulation Tool: MATLAB with Satellite Communications Toolbox
- Orbital Data: MATLAB-generated TLE dataset
- GitHub Repository: <https://github.com/jaehoonpauljeong/IPWASN/tree/main/IETF-126>

Objectives

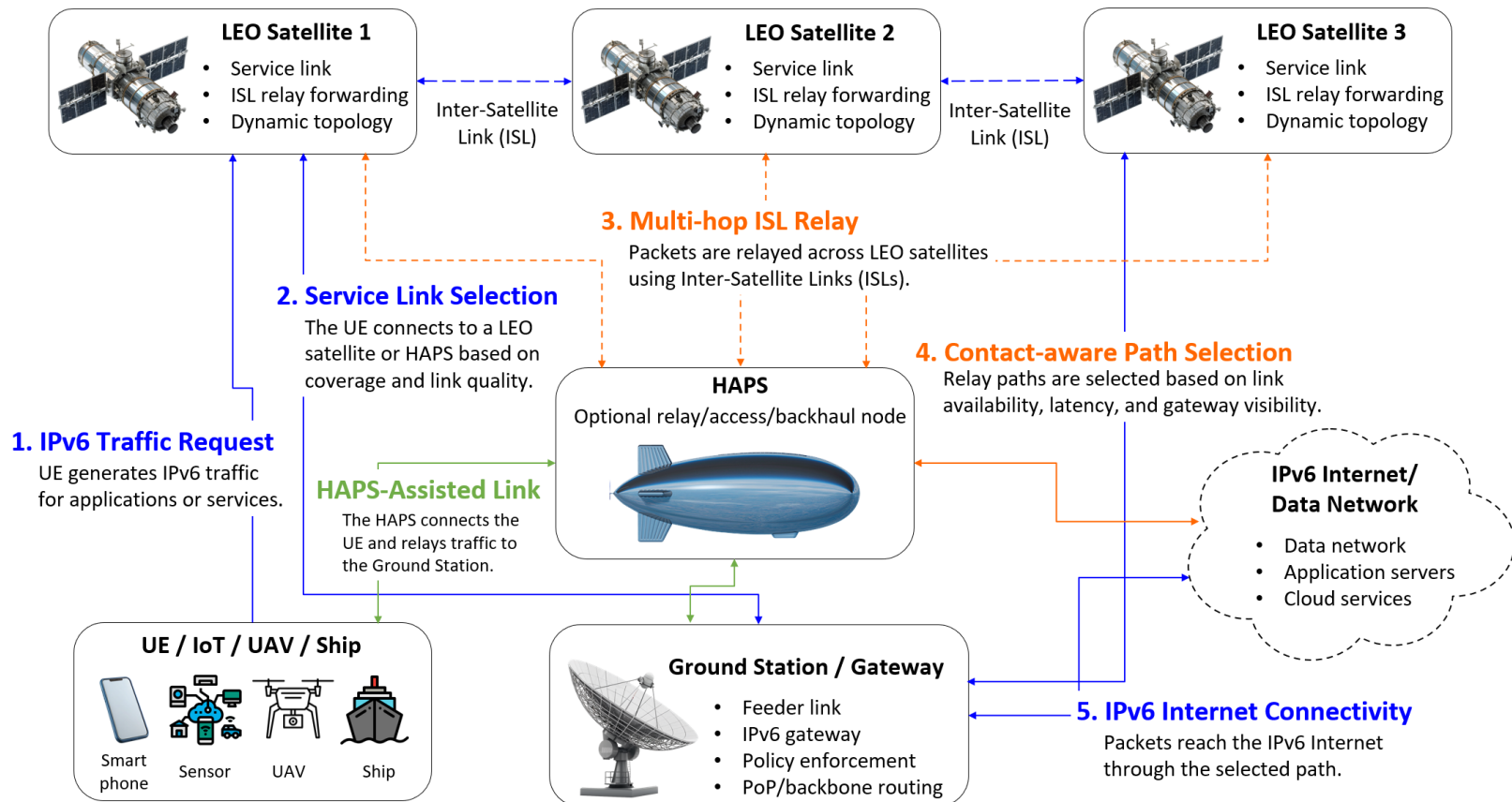
- To define IPWASN for IPv6 connectivity in LEO satellite and HAPS-assisted NTN environments.
- To identify IPv6 requirements for mobile UEs, satellites, HAPS nodes, and gateways.
- To present a reference architecture using services links, inter-satellite links, HAPS relays, and ground gateways.

IPWASN Internet-Draft Structure

<https://datatracker.ietf.org/doc/draft-lee-6man-ipv6-satellite-networks/>

- Introduction: Motivation, scope, and terminology.
- Reference Scenario: IPv6 connectivity across UEs, LEO satellites, HAPS nodes, gateways, and the Internet.
- Problem and Requirements: Mobility, handover, delay, intermittent connectivity, QoS, and scalability.
- Use Cases: Remote access, disaster recovery, maritime/aeronautical communication, UAVs, video, and IoT.
- Security: Protection of Neighbor Discovery, mobility signaling, trajectory/contact data, and privacy.

IPv6 Wireless Access in Satellite Networks (IPWASN) Reference Scenario



Goal of IPWASN Project

- The goal is to identify the problem statement and use cases for IPv6 Wireless Access in Satellite Networks (IPWASN) as Non-Terrestrial Networks (NTNs).
 - **IPWASN (IPv6 Wireless Access in Satellite Networks)**
 - Analyze how IPv6 mechanisms can support ground-controlled data forwarding in satellite NTN environments, focusing on predictable mobility, handover, intermittent connectivity, and Inter-Satellite Link (ISL) relay paths.
 - Outline requirements for SRv6-based source routing, mobility management, QoS support, and reliable data forwarding through ground gateways.
- Internet Draft for the IPWASN Project
 - <https://datatracker.ietf.org/doc/draft-lee-v6ops-ipv6-satellite-networks/>

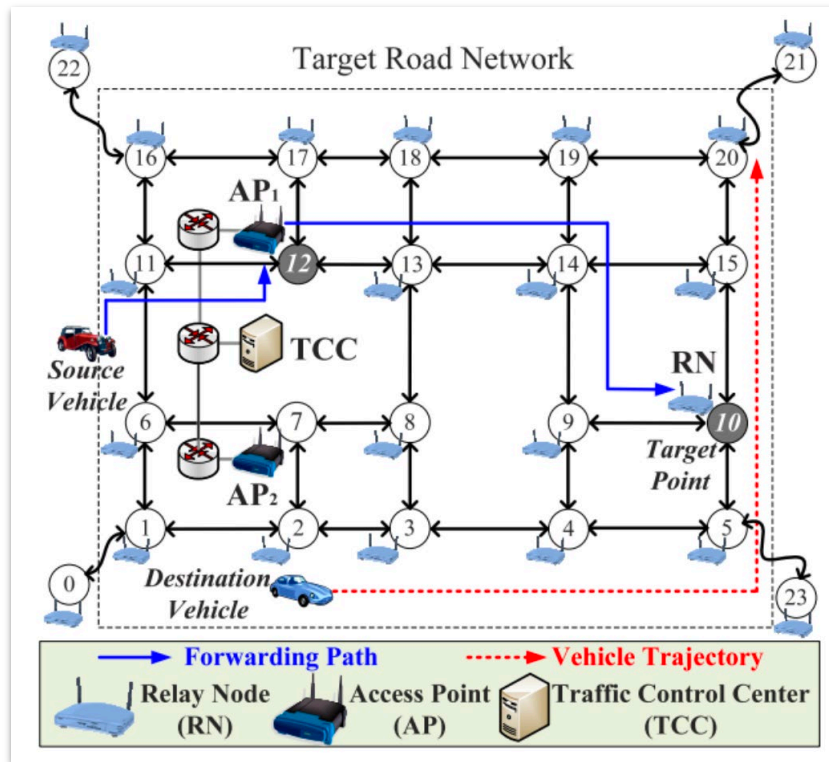
Related Work: Trajectory-Based Statistical Forwarding

- **Relay-Assisted Data Forwarding**

- Step 1: Access Point (AP) → Target Point
 - Traffic Control Center (TCC) selects a target point on the predicted trajectory and source-routes packets to the relay node near that point.
- Step 2: Target Point → Destination Vehicle
 - The relay node buffers or forwards packets depending on the vehicle's arrival timing.

- **Insight for Satellite Networks**

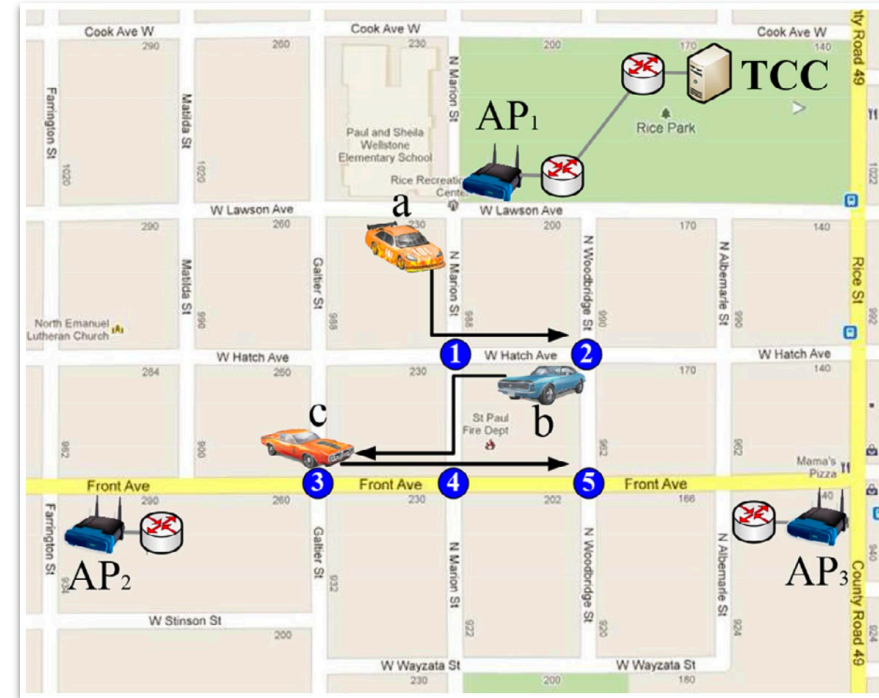
- A satellite as a relay node can carry and forward data packets based on a predicted future contact opportunity.



TSF: Trajectory-Based Statistical Forwarding for V2I

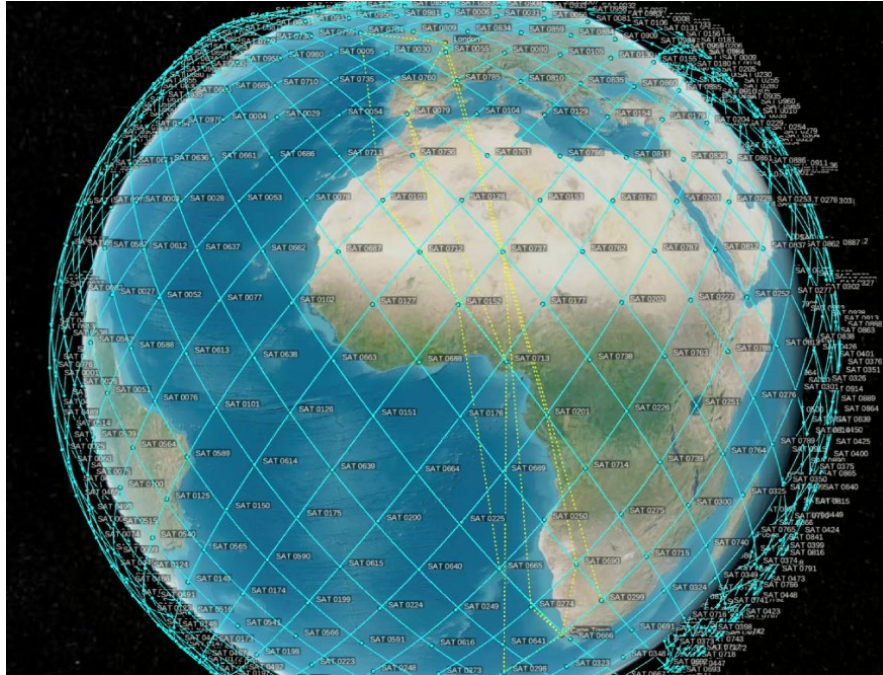
Related Work: Trajectory-Based Data Forwarding

- **Travel Prediction-Based Forwarding Process**
 - Step 1: Predict Future Connectivity
 - The ground controller predicts future satellite contacts from deterministic orbital motion.
 - Step 2: Compare Terrestrial and LEO Paths
 - The controller compares terrestrial fiber delay with LEO relay-path delay over time.
 - Step 3: Select a Stable Forwarding Option
 - Selects the LEO path during stable predicted delay-advantage windows.
- **Insight for Satellite Networks**
 - TPD is reused to select a proactive path, consisting of relay satellites.



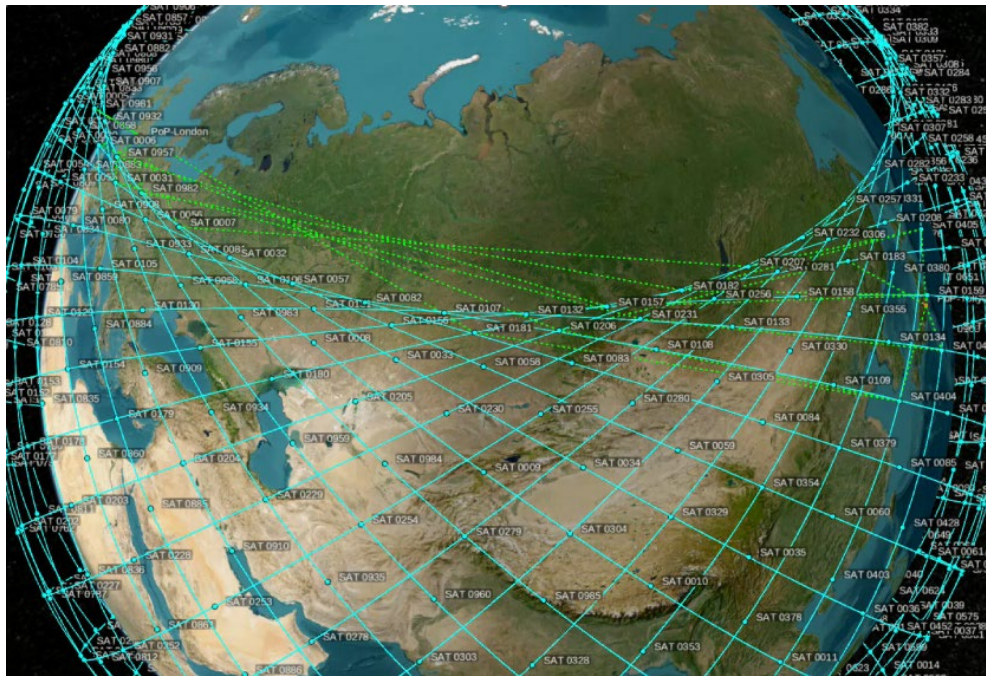
TPD: Travel Prediction-Based Data Forwarding for V2V

Relay Path Simulation in Mega-Constellation



- **Scenario**
 - London → Cape Town over a 20-minute orbital pass
 - 1,000 LEO satellites generated as 40 planes x 25 satellites
- **Connectivity Model**
 - Service link: A ground station can connect to satellites above 10° elevation.
 - Inter-Satellite Link (ISL): Satellites can connect when distance is within 5,000 km.
- **Path Computation**
 - At every 30-second time step, the simulator builds a relay graph from visible satellites and ISLs.
 - The shortest-delay path is selected between the source and destination ground stations.

Prediction-Aware Selection between Terrestrial and LEO Paths



- **Comparison Principle**

- Terrestrial delay = fiber route distance / fiber speed
- LEO delay = feeder link + ISL relay path + feeder link

- **Scenario**

- London Point of Presence (PoP) ↔ Tokyo PoP
- 20-minute orbital pass with 30-second sampling

- **Path Computation**

- Terrestrial path uses 1.4x great-circle distance and $2/3$ c fiber speed.
- LEO path selects the shortest-delay relay path among visible satellites and ISLs within 5,000 km.

Demonstration

```
genlargconste.m x +
/home/jaguar/shared/work/genlargconste.m

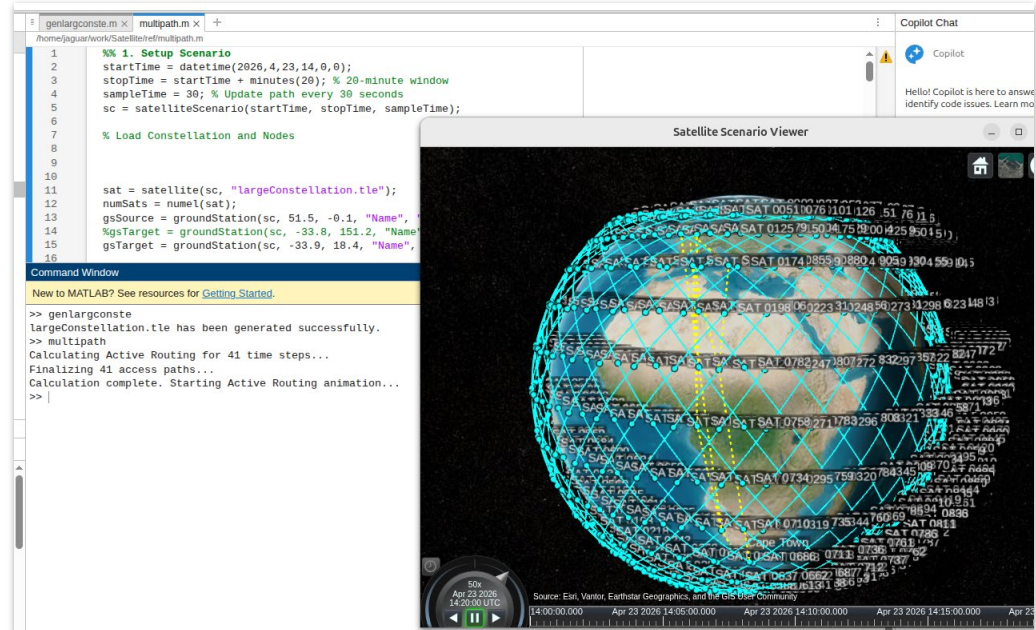
1 % Script to generate a 1000-satellite TLE file (Fixed)
2 fileName = 'largeConstellation.tle';
3 fileID = fopen(fileName, 'w');
4 numPlanes = 40;
5 satsPerPlane = 25;
6 inc = 53.0;
7 alt = 550;
8 mu = 398600.4418; % km^3/s^2
9 Re = 6378.137;
10 % Mean motion in revolutions per day
11 n = (86400) / (2 * pi * sqrt((alt + Re)^3 / mu));
12
13 count = 1;
14 for p = 1:numPlanes
15     raan = (p-1) * (360/numPlanes);
16     for s = 1:satsPerPlane
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
>> genlargconste
largeConstellation.tle has been generated successfully.
>>
```

1. [MATLAB] Run 'genlargconste.m'



2. [MATLAB] Run 'multipath.m'

What we learned

- We identified key challenges for data forwarding over LEO satellite relay networks.
 - LEO satellite forwarding should be modeled as a time-varying relay/contact problem, not only as a dynamic routing problem.
 - Stable forwarding depends on predicted link availability, gateway visibility, path lifetime, and intermittent connectivity.
 - Predictable satellite movement can support contact-aware path computation and scheduled handover preparation.
 - TPD and TSF provide useful design insights for forwarding with predicted mobility and future contact opportunities.

Next Steps

- **Refine IPWASN Problem Statement**

- Clarify satellite relay, ground gateway, and PoP roles in IPv6 satellite networks.
- Model satellites as relay/contact nodes and place IPv6 forwarding decisions at ground gateways and PoPs.

- **TPD-Based Forwarding Analysis**

- Apply TPD/TSF insights to predictable satellite trajectories.
- Compare shortest-delay relay paths with prediction-aware forwarding decisions.

- **IETF-126 Discussion**

- Collect feedback from WG
- Discuss IPv6/SRv6 support for controller-assisted satellite forwarding.

Open-Source Project for IPWASN

[URL] <https://github.com/jaehoonpauljeong/IPWASN/tree/main/IETF-126>

The screenshot shows the GitHub interface for the repository 'jaehoonpauljeong / IPWASN'. The left sidebar displays the file tree with the 'IETF-126' directory selected. The main content area shows the files within this directory, including 'Readme.txt', 'genlargconste.m', 'largeConstellation.tle', and 'multipath.m'. The 'Readme.txt' file is open, showing the text: 'This is the Repository for IPv6 Wireless Access in Satellite Networks.'

Files

main

Go to file

IETF-126

- Readme.txt
- genlargconste.m
- largeConstellation.tle
- multipath.m
- Side-Meetings
- code
- README.md

IPWASN / IETF-126

Add file

jm-lee98 Add files via upload e22ca84 · 14 minutes ago History

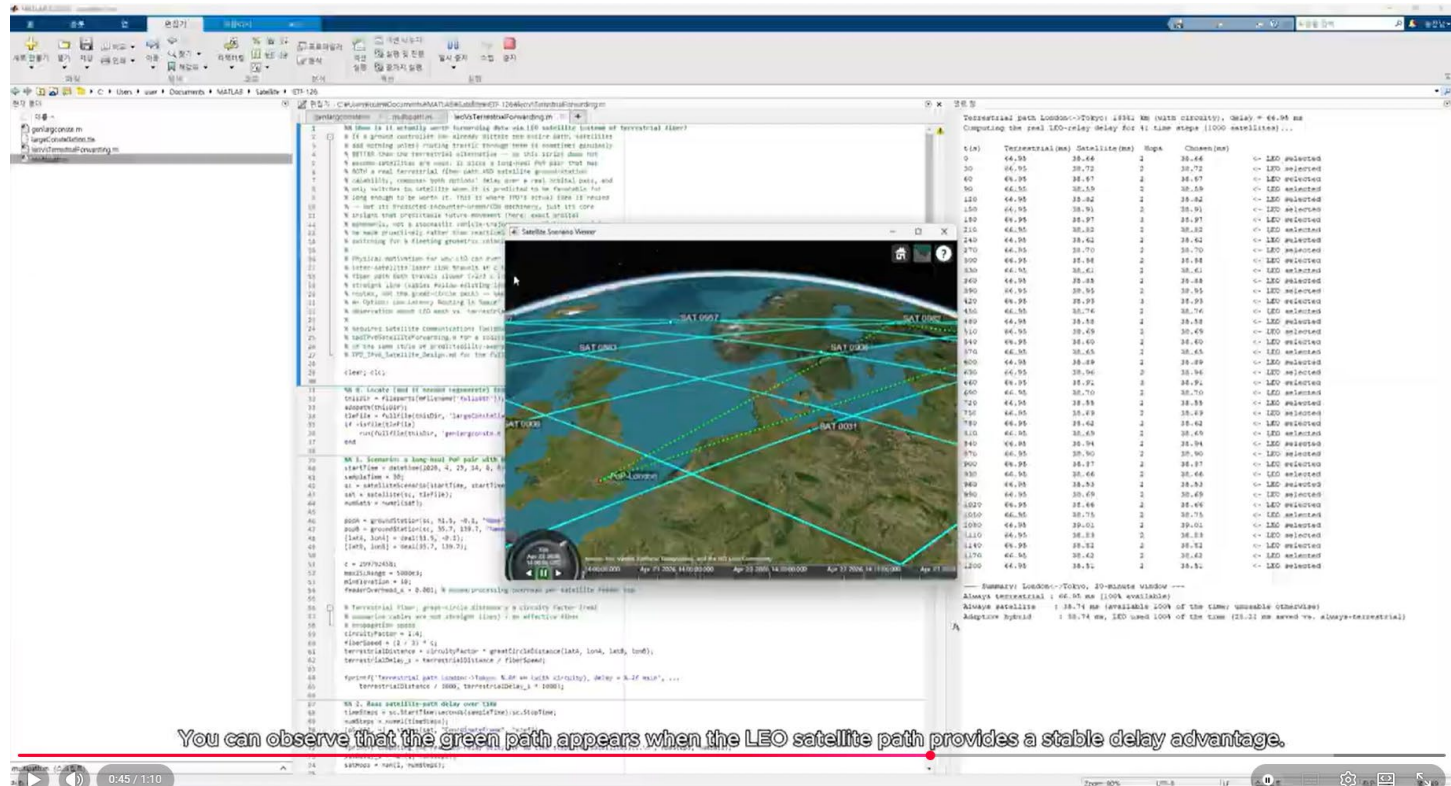
Name	Last commit message	Last commit date
..		
Readme.txt	Create Readme.txt	yesterday
genlargconste.m	Add files via upload	14 minutes ago
largeConstellation.tle	Add files via upload	14 minutes ago
multipath.m	Add files via upload	14 minutes ago

Readme.txt

This is the Repository for IPv6 Wireless Access in Satellite Networks.

Demonstration Video Clip for IPWASN

[URL] https://www.youtube.com/watch?v=3FXah_x0Ri4



You can observe that the green path appears when the LEO satellite path provides a stable delay advantage.

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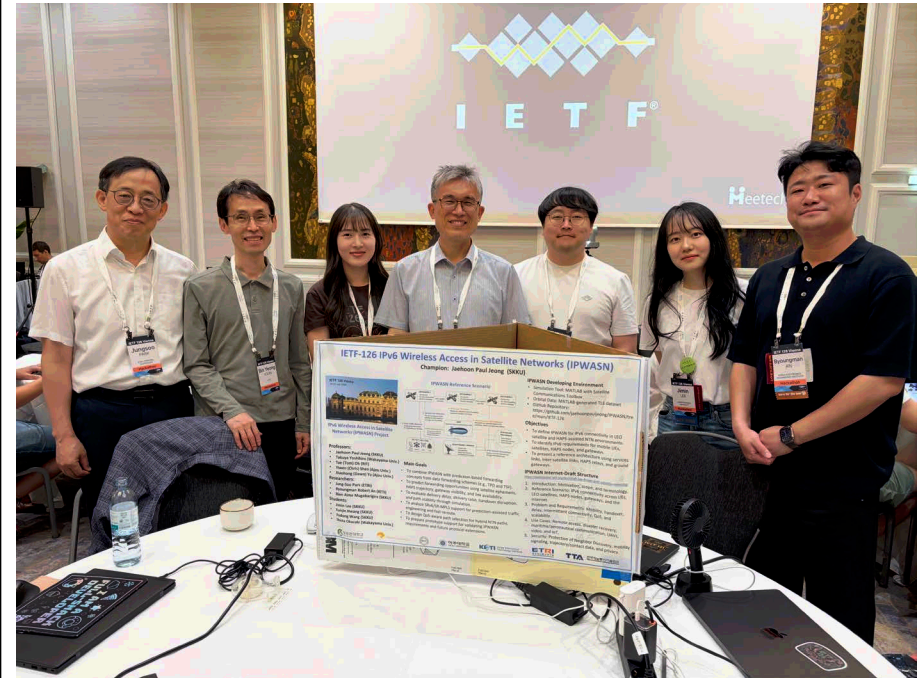
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Team Photo



Thank You
and
Any Comments?